1.09 Standard Series

Worked Solutions

1 a)

$$\sum_{r=1}^{30} 8 = 8 \times 30 = 240$$

1 b)

$$\sum_{r=1}^{50} r = \frac{1}{2} \times 50(50+1) = 1275$$

1 c)

$$\sum_{r=1}^{20} r^2 = \frac{1}{6} \times 20(20+1)(40+1) = 2870$$

1 d)

$$\sum_{r=1}^{10} r^3 = \frac{1}{4} \times 10^2(10+1)^2 = 3025$$

2 a)

$$\sum_{r=1}^{n}(2r+3) = 2\sum_{r=1}^{n} r + 3\sum_{r=1}^{n} 1$$
$$= 2 \times \frac{1}{2}n(n+1) + 3n = n^2 + 4n$$

2 b)

$$\sum_{r=1}^{n}(5r-12) = 5\sum_{r=1}^{n} r - 12\sum_{r=1}^{n} 1$$
$$= 5 \times \frac{1}{2}n(n+1) - 12n = \frac{5}{2}n^2 - \frac{19}{2}n$$

2 c)

$$\sum_{r=1}^{n}(4r+1) = 4\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1$$
$$= 4 \times \frac{1}{2}n(n+1) + n = 2n^2 + 3n$$

2 d)

$$\sum_{r=1}^{n}(ar+b) = a\sum_{r=1}^{n} r + b\sum_{r=1}^{n} 1$$
$$= \frac{1}{2}an(n+1) + bn = \frac{1}{2}an^2 + \frac{1}{2}an + bn$$

3)

$$\sum_{r=1}^{k}(2r+k) = 2\sum_{r=1}^{k} r + k\sum_{r=1}^{k} 1$$
$$= 2 \times \frac{1}{2}k(k+1) + k^2 = 2k^2 + k$$

$2k^2 + k = 300 \Rightarrow 2k^2 + k - 300 = 0$ which gives $k = -\frac{25}{2}$ and $k = 12$

$$\therefore k = 12$$

4)

$$\sum_{r=1}^{12} r(6r-1) = 6\sum_{r=1}^{12} r^2 - \sum_{r=1}^{12} r$$
$$= 6 \times \frac{1}{6} \times 12(12+1)(24+1) - \frac{1}{2} \times 12 \times (12+1)$$
$$= 3822 \text{ as req.}$$

5)

$$\sum_{r=1}^{50}(3r+2)^2 = \sum_{r=1}^{50}(9r^2 + 12r + 4)$$
$$= 9\sum_{r=1}^{50} r^2 + 12\sum_{r=1}^{50} r + 4\sum_{r=1}^{50} 1$$
$$= 9 \times \frac{1}{6} \times 50(50+1)(100+1) + 12 \times \frac{1}{2} \times 50 \times 51 + 4 \times 50$$
$$= 401\,825$$

6 a)

$$n(n+1) + 4n$$
$$= n[n+1+4]$$
$$= n(n+5)$$

6 b)

$$\begin{aligned}
&\frac{7}{2}n(n+1) - 5n \\
&= \frac{1}{2}n[7(n+1) - 10] \\
&= \frac{1}{2}n[7n + 7 - 10] \\
&= \frac{1}{2}n(7n - 3)
\end{aligned}$$

6 c)

$$\begin{aligned}
&3n(n+1)(2n+1) + \frac{1}{2}n(n+1) \\
&= \frac{1}{2}n(n+1)[6(2n+1) + 1] \\
&= \frac{1}{2}n(n+1)(12n+7)
\end{aligned}$$

6 d)

$$\begin{aligned}
&4n(n+1)(2n+1) - \frac{1}{4}n^2(n+1)^2 \\
&= \frac{1}{4}n(n+1)[16(2n+1) - n(n+1)] \\
&= \frac{1}{4}n(n+1)[32n + 16 - n^2 - n] \\
&= \frac{1}{4}n(n+1)[-n^2 + 31n + 16]
\end{aligned}$$

6 e)

$$\begin{aligned}
&\frac{1}{2}n^2(n+1)^2 - 3n(n+1) \\
&= \frac{1}{2}n(n+1)[n(n+1) - 6] \\
&= \frac{1}{2}n(n+1)[n^2 + n - 6] \\
&= \frac{1}{2}n(n+1)(n+3)(n-2)
\end{aligned}$$

6 f)

$$\begin{aligned}
&\frac{1}{4}n^2(n+1)^2 - \frac{1}{6}n(n+1)(2n+1) \\
&= \frac{1}{12}n(n+1)[3n(n+1) - 2(2n+1)] \\
&= \frac{1}{12}(n+1)[3n^2 + 3n - 4n - 2] \\
&= \frac{1}{12}(n+1)[3n^2 - n - 2] \\
&= \frac{1}{12}(n+1)(3n+2)(n-1)
\end{aligned}$$

6 g)

$$\begin{aligned}
&\frac{3}{4}n^2(n+1)^2 + \frac{5}{2}n(n+1) \\
&= \frac{1}{4}n(n+1)[3n(n+1)+10] \\
&= \frac{1}{4}n(n+1)(3n^2+3n+10)
\end{aligned}$$

7 a)

$$\begin{aligned}
\sum_{r=1}^{n}(r^3+r^2) &= \frac{1}{4}n^2(n+1)^2 + \frac{1}{6}n(n+1)(2n+1) \\
&= \frac{1}{12}n(n+1)[3n(n+1)+2(2n+1)] \\
&= \frac{1}{12}n(n+1)[3n^2+3n+4n+2] \\
&= \frac{1}{12}n(n+1)(3n^2+7n+2) \\
&= \frac{1}{12}n(n+1)(3n+1)(n+2)
\end{aligned}$$

7 b)

$$\begin{aligned}
\sum_{r=1}^{n}(2r^2+5r) &= 2\sum_{r=1}^{n}r^2 + 5\sum_{r=1}^{n}r \\
&= 2\times\frac{1}{6}n(n+1)(2n+1) + 5\times\frac{1}{2}n(n+1) \\
&= \frac{1}{6}n(n+1)[2(2n+1)+15] \\
&= \frac{1}{6}n(n+1)(4n+17)
\end{aligned}$$

7 c)

$$\begin{aligned}
\sum_{r=1}^{n}(7r^3-2r^2) &= 7\sum_{r=1}^{n}r^3 - 2\sum_{r=1}^{n}r^2 \\
&= 7\times\frac{1}{4}n^2(n+1) - 2\times\frac{1}{6}n(n+1)(2n+1) \\
&= \frac{1}{12}n(n+1)[21n(n+1)-4(2n+1)] \\
&= \frac{1}{12}n(n+1)[21n^2+21n-8n-4] \\
&= \frac{1}{12}n(n+1)(21n^2+13n-4)
\end{aligned}$$

7 d)

$$\sum_{r=1}^{n}(1-2r+5r^3) = \sum_{r=1}^{n}1 - 2\sum_{r=1}^{n}r + 5\sum_{r=1}^{n}r^3$$
$$= n - 2\times\frac{1}{2}n(n+1) + 5\times\frac{1}{4}n^2(n+1)^2$$
$$= \frac{1}{4}n[4-4(n+1)+5n(n+1)^2]$$
$$= \frac{1}{4}n[4-4n-4+5n(n^2+2n+1)]$$
$$= \frac{1}{4}n[-4n+5n^3+10n^2+5n]$$
$$= \frac{1}{4}n^2(5n^2+10n+1)$$

7 e)

$$\sum_{r=1}^{n}r(2r+1) = 2\sum_{r=1}^{n}r^2 + \sum_{r=1}^{n}r$$
$$= 2\times\frac{1}{6}n(n+1)(2n+1) + \frac{1}{2}n(n+1)$$
$$= \frac{1}{6}n(n+1)[2(2n+1)+3]$$
$$= \frac{1}{6}n(n+1)(4n+5)$$

7 f)

$$\sum_{r=1}^{n}r^2(r-4) = \sum_{r=1}^{n}r^3 - 4\sum_{r=1}^{n}r^2$$
$$= \frac{1}{4}n^2(n+1)^2 - 4\times\frac{1}{6}n(n+1)(2n+1)$$
$$= \frac{1}{12}n(n+1)[3n(n+1)-8(2n+1)]$$
$$= \frac{1}{12}n(n+1)[3n^2+3n-16n-8]$$
$$= \frac{1}{12}n(n+1)(3n^2-13n-8)$$

8)

$$\sum_{r=1}^{n}(4r-2)^2 = \sum_{r=1}^{n}(16r^2-16r+4)$$
$$= 16\sum_{r=1}^{n}r^2 - 16\sum_{r=1}^{n}r + 4\sum_{r=1}^{n}1$$
$$= 4\left[4\sum_{r=1}^{n}r^2 - 4\sum_{r=1}^{n}r + \sum_{r=1}^{n}1\right]$$
$$= 4\left[4\times\frac{1}{6}n(n+1)(2n+1) - 4\times\frac{1}{2}n(n+1) + n\right]$$
$$= \frac{4}{3}n[2(n+1)(2n+1) - 6(n+1) + 3]$$
$$= \frac{4}{3}n[2(2n^2+3n+1) - 6(n+1) + 3]$$
$$= \frac{4}{3}n[4n^2+6n+2-6n-6+3]$$
$$= \frac{4}{3}n[4n^2-1] = \frac{4}{3}n(2n-1)(2n+1)$$

9 a)

$$(1\times2)+(2\times3)+(3\times4)+\cdots = \sum_{r=1}^{n}r(r+1)$$
$$= \sum_{r=1}^{n}r^2 + \sum_{r=1}^{n}r$$
$$= \frac{1}{6}n(n+1)(2n+1) + \frac{1}{2}n(n+1)$$
$$= \frac{1}{6}n(n+1)[2n+1+3]$$
$$= \frac{1}{3}n(n+1)(n+2)$$

9 b)

$$(3\times5)+(4\times6)+(5\times7)+\cdots = \sum_{r=1}^{n}(r+2)(r+4)$$
$$= \sum_{r=1}^{n}(r^2+6r+8)$$
$$= \sum_{r=1}^{n}r^2 + 6\sum_{r=1}^{n}r + 8\sum_{r=1}^{n}1$$
$$= \frac{1}{6}n(n+1)(2n+1) + 6\times\frac{1}{2}n(n+1) + 8n$$
$$= \frac{1}{6}n[(n+1)(2n+1) + 18(n+1) + 48]$$
$$= \frac{1}{6}n[2n^2+3n+1+18n+18+48]$$
$$= \frac{1}{6}n(2n^2+21n+67)$$

9 c)

$$(2 \times 4) + (3 \times 5) + (4 \times 6) + \cdots = \sum_{r=1}^{n} (r+1)(r+3)$$

$$= \sum_{r=1}^{n} (r^2 + 4r + 3)$$

$$= \sum_{r=1}^{n} r^2 + 4\sum_{r=1}^{n} r + 3\sum_{r=1}^{n} 1$$

$$= \frac{1}{6}n(n+1)(2n+1) + 4 \times \frac{1}{2}n(n+1) + 3n$$

$$= \frac{1}{6}n[(n+1)(2n+1) + 12(n+1) + 18]$$

$$= \frac{1}{6}n[2n^2 + 3n + 1 + 12n + 12 + 18]$$

$$= \frac{1}{6}n(2n^2 + 15n + 31)$$

9 d)

$$2 \times 1^2 + 3 \times 2^2 + 4 \times 3^2 + \cdots = \sum_{r=1}^{n} (r+1)r^2$$

$$= \sum_{r=1}^{n} (r^3 + r^2)$$

$$= \sum_{r=1}^{n} r^3 + \sum_{r=1}^{n} r^2$$

$$= \frac{1}{4}n^2(n+1)^2 + \frac{1}{6}n(n+1)(2n+1)$$

$$= \frac{1}{12}n(n+1)[3n(n+1) + 2(2n+1)]$$

$$= \frac{1}{12}n(n+1)[3n^2 + 3n + 4n + 2]$$

$$= \frac{1}{12}n(n+1)[3n^2 + 7n + 2]$$

$$= \frac{1}{12}n(n+1)(3n+1)(n+2)$$

9 e)

$$
\begin{aligned}
&(1\times 3\times 4)+(2\times 4\times 5)+(3\times 5\times 6)+\cdots\\
&=\sum_{r=1}^{n} r(r+2)(r+3)\\
&=\sum_{r=1}^{n}(r^3+5r^2+6r)\\
&=\sum_{r=1}^{n} r^3+5\sum_{r=1}^{n} r^2+6\sum_{r=1}^{n} r\\
&=\frac{1}{4}n^2(n+1)^2+5\times\frac{1}{6}n(n+1)(2n+1)+6\times\frac{1}{2}n(n+1)\\
&=\frac{1}{12}n(n+1)[3n(n+1)+10(2n+1)+36]\\
&=\frac{1}{12}n(n+1)[3n^2+3n+20n+10+36]\\
&=\frac{1}{12}n(n+1)(3n^2+23n+46)
\end{aligned}
$$

10 a)

$$
\begin{aligned}
\sum_{r=1}^{n}(r+1)(3r-1)&=\sum_{r=1}^{n}(3r^2+2r-1)\\
&=3\sum_{r=1}^{n} r^2+2\sum_{r=1}^{n} r-\sum_{r=1}^{n} 1\\
&=3\times\frac{1}{6}n(n+1)(2n+1)+2\times\frac{1}{2}n(n+1)-n\\
&=\frac{1}{2}n[(n+1)(2n+1)+2(n+1)-2]\\
&=\frac{1}{2}n[2n^2+3n+1+2n+2-2]\\
&=\frac{1}{2}n(2n^2+5n+1)
\end{aligned}
$$

10 b)

$2\times 2+3\times 5+4\times 8+\cdots+51\times 149$ is $\sum_{r=1}^{n}(r+1)(3r-1)$ with $r=1$ and $n=50$

$$\frac{1}{2}n(2n^2+5n+1)=\frac{1}{2}\times 50(2\times 50^2+5\times 50+1)=131\,275$$

11 a)

$$\sum_{r=1}^{2n} r^2=\frac{1}{6}(2n)(2n+1)(2(2n)+1)=\frac{1}{3}n(2n+1)(4n+1)$$

11 b)

$$
\begin{aligned}
\sum_{r=1}^{2n}(2r)^2&=4\sum_{r=1}^{2n} r^2\\
&=\frac{4}{6}(2n)(2n+1)(2(2n)+1)=\frac{4}{3}n(2n+1)(4n+1)
\end{aligned}
$$

11 c)

$$\sum_{r=1}^{2n}(2r-1)^2 = 4\sum_{r=1}^{2n} r^2 - 4\sum_{r=1}^{2n} r + \sum_{r=1}^{2n} 1$$
$$= \frac{4}{6}(2n)(2n+1)(2(2n)+1) - 4\times\frac{1}{2}(2n)(2n+1) + 2n$$
$$= \frac{4}{3}n(2n+1)(4n+1) - 4n(2n+1) + 2n$$
$$= \frac{4}{3}n(2n+1)(4n+1) - 8n^2 - 2n$$
$$= \frac{4}{3}n(2n+1)(4n+1) - 2n(4n+1)$$
$$= \frac{2}{3}n(4n+1)[2(2n+1)-3]$$
$$= \frac{2}{3}(4n+1)(4n-1)$$

12 a)

$$2^2 + 4^2 + 6^2 + \cdots + 20^2 = \sum_{r=1}^{10}(2r)^2 = 4\sum_{r=1}^{10} r^2$$
$$= 4\times\frac{1}{6}\times 10\times 11\times 21$$
$$= 1540 \text{ as req.}$$

12 b)

$$1^2 + 3^2 + 5^2 + \cdots + 19^2 = \sum_{r=1}^{10}(2r-1)^2 = \sum_{r=1}^{10}(4r^2 - 4r + 1)$$
$$= 4\sum_{r=1}^{10} r^2 - 4\sum_{r=1}^{10} r + \sum_{r=1}^{10} 1$$
$$= 4\times\frac{1}{6}\times 10\times 11\times 21 - 4\times\frac{1}{2}\times 10\times 11 + 10$$
$$= 1330 \text{ as req.}$$

12 c)

$$2^3 + 4^3 + 6^3 + \cdots + 20^3 = \sum_{r=1}^{10}(2r)^3 = 8\sum_{r=1}^{10} r^3$$
$$= 8\times\frac{1}{4}\times 10^2\times 11^2$$
$$= 24\,200 \text{ as req.}$$

13)

$$1 + 2 + 3 + 4 + 5 + \cdots = \sum_{r=1}^{999} r = \frac{1}{2}\times 999(999+1) = 499\,500$$

The square numbers up to 999 are $1^2 + 2^2 + 3^2 + \cdots + 31^2$

$$1^2 + 2^2 + 3^2 + \cdots + 31^2 = \sum_{r=1}^{31} r^2 = \frac{1}{6} \times 31(31+1)(62+1) = 10\,416$$

$$499\,500 - 10\,416 = 489\,084$$

14)

$2^2 + 4^2 + 6^2 + \cdots + (2r)^2 = 82160$ where the series has n terms.

$$\sum_{r=1}^{n} (2r)^2 = 4\sum_{r=1}^{n} r^2$$
$$= 4 \times \frac{1}{6} n(n+1)(2n+1)$$
$$\frac{2}{3} n(n+1)(2n+1) = 82160$$
$$n(n+1)(2n+1) = 123240$$
$$2n^3 + 3n^2 + n = 123240$$
$$2n^3 + 3n^2 + n - 123240 = 0$$

which has the single real solution of $n = 39$

15)

$1^2 + 3^2 + 5^2 + \cdots + (2r-1)^2 = 20\,825$ where the series has n terms.

$$\sum_{r=1}^{n} (2r-1)^2 = \sum_{r=1}^{n} (4r^2 - 4r + 1)$$
$$= 4\sum_{r=1}^{n} r^2 - 4\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1$$
$$= 4 \times \frac{1}{6} n(n+1)(2n+1) - 4 \times \frac{1}{2} n(n+1) + n$$
$$\frac{2}{3} n(n+1)(2n+1) - 2n(n+1) + n = 20\,825$$
$$\frac{2}{3} n(2n^2 + 3n + 1) - 2n^2 - 2n + n = 20\,825$$
$$2n(2n^2 + 3n + 1) - 6n^2 - 6n + 3n = 62\,475$$
$$4n^3 + 6n^2 + 2n - 6n^2 - 6n + 3n = 62\,475$$
$$4n^3 - n = 62\,475$$
$$4n^3 - n - 62\,475 = 0$$

which has the single real solution of $n = 25$

16 a)

$$\sum_{r=1}^{n}(r+2)(r+3) = \sum_{r=1}^{n}(r^2+5r+6)$$
$$= \sum_{r=1}^{n} r^2 + 5\sum_{r=1}^{n} r + 6\sum_{r=1}^{n} 1$$
$$= \frac{1}{6}n(n+1)(2n+1) + 5 \times \frac{1}{2}n(n+1) + 6n$$
$$24 + 3\sum_{r=1}^{n}(r+2)(r+3)$$
$$= 24 + \frac{1}{2}n(n+1)(2n+1) + \frac{15}{2}n(n+1) + 18n$$
$$= 24 + \frac{1}{2}n(n+1)(2n+1) + \frac{15}{2}n(n+1) + 18n$$
$$= 24 + n^3 + \frac{3}{2}n^2 + \frac{1}{2}n + \frac{15}{2}n^2 + \frac{15}{2}n + 18n$$
$$= n^3 + 9n^2 + 26n + 24$$
$$= (n+2)(n^2+7n+12)$$
$$= (n+2)(n+3)(n+4)$$

16 b)

$$24 + 3\sum_{r=1}^{n}(r+2)(r+3) = (n+2)(n+3)(n+4)$$

which are consecutive integers, one of which must be divisible by 2 and another by 3 therefore $24 + 3\sum_{r=1}^{n}(r+2)(r+3)$ is divisible by 6

17 a)

$$\sum_{r=1}^{n}(7r-2)^2 = \sum_{r=1}^{n}(49r^2-28r+4)$$
$$= 49\sum_{r=1}^{n} r^2 - 28\sum_{r=1}^{n} r + 4\sum_{r=1}^{n} 1$$
$$= 49 \times \frac{1}{6}n(n+1)(2n+1) - 28 \times \frac{1}{2}n(n+1) + 4n$$
$$= \frac{1}{6}n[49(n+1)(2n+1) - 84(n+1) + 24]$$
$$= \frac{1}{6}n[49(2n^2+3n+1) - 84n - 84 + 24]$$
$$= \frac{1}{6}n[98n^2+147n+49-84n-84+24]$$
$$= \frac{1}{6}n(98n^2+63n-11)$$

$a = 98$, $b = 63$ and $c = -11$

17 b)

$$\frac{1}{6}k(98k^2+63k-11)=190k^2$$
$$98k^3+63k^2-11k=1140k^2$$
$$98k^3-1077k^2-11k=0$$

which has solutions $k=0$, $k=-\frac{1}{98}$ and $k=11$

$$\therefore k=11$$

18 a)

$$\sum_{r=1}^{n} ar+b = a\sum_{r=1}^{n} r+\sum_{r=1}^{n} b = a\times\frac{1}{2}n(n+1)+bn$$
$$a\times\frac{1}{2}\times 3\times 4+3b=0\Rightarrow 6a+3b=0$$
$$a\times\frac{1}{2}\times 7\times 8+7b=35\Rightarrow 28a+7b=35$$

Gives $a=\frac{5}{2}$ and $b=-5$
So that $\mathrm{f}(r)=\frac{5}{2}r-5$

18 b)

$$\sum_{r=1}^{20}\frac{5}{2}r-5=\frac{5}{2}\times\frac{1}{2}\times 20\times 21-5\times 20=425$$

19 a)

$$\sum_{r=1}^{n}(5r-7)=5\sum_{r=1}^{n} r-\sum_{r=1}^{n} 7=5\times\frac{1}{2}n(n+1)-7n$$
$$=\frac{1}{2}n(5(n+1)-14)$$
$$=\frac{1}{2}n(5n+5-14)$$
$$=\frac{1}{2}n(5n-9)\text{ as req.}$$

19 b)

$$\frac{1}{2}n(5n-9)>6000$$
$$5n^2-9n-12000>0$$

$5n^2-9n-12000=0$ has solutions $n=\frac{9}{10}\pm\frac{1}{10}\sqrt{240081}$
So $n=50$

20)

$$\sum_{r=1}^{n}(ar+b) = a\sum_{r=1}^{n} r + bn$$
$$= a \times \frac{1}{2}n(n+1) + bn$$
$$= \frac{1}{2}an^2 + \frac{1}{2}an + bn$$
$$= \frac{1}{2}an^2 + \left(\frac{1}{2}a + b\right)n$$

$\frac{1}{2}a = 1$ so $a = 2$

$$\frac{1}{2} \times 2 + b = -10$$

$\therefore a = 2$ and $b = -11$

21 a)

$$\sum_{r=1}^{n}(r+10)(r+12) = \sum_{r=1}^{n}(r^2 + 22r + 120)$$
$$= \sum_{r=1}^{n} r^2 + 22\sum_{r=1}^{n} r + 120\sum_{r=1}^{n} 1$$
$$= \frac{1}{6}n(n+1)(2n+1) + 22 \times \frac{1}{2}n(n+1) + 120n$$
$$= \frac{1}{6}n[(n+1)(2n+1) + 66(n+1) + 720]$$
$$= \frac{1}{6}n[2n^2 + 3n + 1 + 66n + 66 + 720]$$
$$= \frac{1}{6}n(2n^2 + 69n + 787) \text{ as req.}$$

21 b)

$$\frac{1}{6}n(2n^2 + 69n + 787) = 3806$$
$$2n^3 + 69n^2 + 787n = 22836$$
$$2n^3 + 69n^2 + 787n - 22836 = 0$$

The only real solution is $n = 12$

22 a)

$$210 + 40r$$

22 b)

$$\left(50 + \frac{r^2}{30}\right)(210 + 40r) - (2000 + 7r^2)$$
$$= 10500 + 2000r + 7r^2 + \frac{4}{3}r^3 - 2000 - 7r^2$$
$$= 8500 + 2000r + \frac{4}{3}r^3$$
$$= \frac{4}{3}(r^3 + 1500r + 6375) \text{ as req.}$$

22 c)

$$\sum_{r=1}^{n} \frac{4}{3}(r^3 + 1500r + 6375)$$
$$= \frac{4}{3}\sum_{r=1}^{n} r^3 + 2000\sum_{r=1}^{n} r + 8500\sum_{r=1}^{n} 1$$
$$= \frac{4}{3} \times \frac{1}{4}n^2(n+1)^2 + 2000 \times \frac{1}{2}n(n+1) + 8500n$$
$$= \frac{1}{3}n[n(n+1)^2 + 3000(n+1) + 25500]$$
$$= \frac{1}{3}n[n(n^2 + 2n + 1) + 3000n + 3000 + 25500]$$
$$= \frac{1}{3}n(n^3 + 2n^2 + 3001n + 28500)$$

22 d)

$$\frac{1}{3} \times 15(15^3 + 2 \times 15^2 + 3001 \times 15 + 28500) = £386\ 700$$

This result is much higher than the actual profit, so the model is over-estimating the profit and so it is not a good model.

23 a)

$$397 + 3r$$

23 b)

$$\sum_{r=1}^{n} 397 + 3r = 397\sum_{r=1}^{n} 1 + 3\sum_{r=1}^{n} r$$
$$= 397n + 3 \times \frac{1}{2}n(n+1)$$
$$= \frac{1}{2}n[794 + 3(n+1)]$$
$$= \frac{1}{2}n(3n + 797)$$

There are n loaves so that $\frac{1}{2}(3n + 797)$ as req.

23 c)

Each loaf has mass $(397 + 3r)$

$$\sum_{r=1}^{20}(397 + 3r) = \frac{1}{2} \times 20(3 \times 20 + 797) = 8570$$

$$\sum_{r=1}^{20}(397 + 3r)^2 = \sum_{r=1}^{20}(9r^2 + 2382r + 157609)$$

$$= 9\sum_{r=1}^{20} r^2 + 2832\sum_{r=1}^{20} r + 157609\sum_{r=1}^{20} 1$$

$$= 9 \times \frac{1}{6} \times 20(20 + 1)(40 + 1) + 2382 \times \frac{1}{2} \times 20 \times 21 + 20 \times 157609$$

$$= 3678230$$

$$\sqrt{\frac{\sum r^2}{n} - \left(\frac{\sum r}{n}\right)^2} = \sqrt{\frac{3678230}{20} - \left(\frac{8570}{20}\right)^2}$$

$$= \sqrt{183911.5 - 428.5^2} = 17.3 \text{ grams (3 SF)}$$

24 a)

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n + 1)^2$$

$$1 + 2 + 3 + \cdots + n = \sum_{r=1}^{n} r = \frac{1}{2}n(n + 1)$$

$$\frac{1^3 + 2^3 + 3^3 + \cdots + n^3}{1 + 2 + 3 + \cdots + n} = \frac{\frac{1}{4}n^2(n + 1)^2}{\frac{1}{2}n(n + 1)} = \frac{1}{2}n(n + 1) \text{ as req.}$$

24 b)

$$\frac{1}{2}x(x + 1) > 200$$

$$\frac{1}{2}x^2 + \frac{1}{2}x - 200 > 0$$

$\frac{1}{2}x^2 + \frac{1}{2}x - 200 = 0$ has roots $x = -\frac{1}{2} \pm \frac{1}{2}\sqrt{1601}$

$$x = 19.506\ldots$$

$$\therefore n = 20$$

25 a)

$$\sum_{r=1}^{n}(r^3 - 2r) = \sum_{r=1}^{n} r^3 - 2\sum_{r=1}^{n} r$$

$$= \frac{1}{4}n^2(n + 1)^2 - 2 \times \frac{1}{2}n(n + 1)$$

$$= \frac{1}{4}n(n + 1)[n(n + 1) - 4]$$

$$= \frac{1}{4}n(n + 1)(n^2 + n - 4)$$

25 b)

$$\sum_{r=1}^{k}(2r^3+3r)-\sum_{r=1}^{k}(r^3+5r)=\sum_{r=1}^{k}(r^3-2r)$$
$$=\frac{1}{4}n(n+1)(n^2+n-4)$$
$$\frac{1}{4}n(n+1)(n^2+n-4)=1224$$
$$n(n^3+2n^2-3n-4)=4896$$
$$n^4+2n^3-3n^2-4n-4896=0$$

which has real solutions $n=-9$ and $n=8 \therefore k=8$

26 a)

$$\sum_{r=10-9}^{30-9}((r+9)-15)((r+9)-16)=\sum_{r=1}^{21}(r-6)(r-7)$$

26 b)

$$\sum_{r=4-3}^{16-3}((r+3)^2+3(r+3))=\sum_{r=1}^{13}(r^2+9r+18)$$

26 c)

$$\sum_{r=12-11}^{20-11}(3(r+11)^3-1)=\sum_{r=1}^{9}(3(r+11)^3-1)$$
$$=\sum_{r=1}^{9}(3(r^3+33r^2+363r+1331)-1)$$
$$=\sum_{r=1}^{9}(3r^3+99r^2+1089r+3992)$$

26 d)

$$\sum_{r=1}^{8}r^2-\sum_{r=1}^{4}r^2=\sum_{r=1}^{4}r^2+\sum_{r=5}^{8}r^2-\sum_{r=1}^{4}r^2$$
$$=\sum_{r=5}^{8}r^2=\sum_{r=5-4}^{8-4}(r+4)^2=\sum_{r=1}^{4}(r+4)^2$$
$$=\sum_{r=1}^{4}(r^2+8r+16)$$

27 a)

$$\sum_{r=10}^{20}(r^2-3)=\sum_{r=1}^{11}((r+9)^2-3)=\sum_{r=1}^{11}(r^2+18r+78)$$
$$=\sum_{r=1}^{11}r^2+18\sum_{r=1}^{11}r+78\sum_{r=1}^{11}1$$
$$=\frac{1}{6}\times 11(23)(12)+18\times\frac{11}{2}(12)+78\times 11$$
$$=2552$$

27 b)

$$\sum_{r=5}^{10}r(r-1)=\sum_{r=1}^{6}(r+4)(r+3)=\sum_{r=1}^{6}(r^2+7r+12)$$
$$=\sum_{r=1}^{6}r^2+7\sum_{r=1}^{6}r+12\sum_{r=1}^{6}1$$
$$=\frac{1}{6}\times 6(13)(7)+7\times\frac{6}{2}(7)+12\times 6$$
$$=310$$

27 c)

$$\sum_{r=50}^{100}(r^3-1)=\sum_{r=1}^{51}((r+49)^3-1)$$
$$=\sum_{r=1}^{51}(r^3+147r^2+7203r+117648)$$
$$=\sum_{r=1}^{51}r^3+147\sum_{r=1}^{51}r^2+7203\sum_{r=1}^{51}r+117648\sum_{r=1}^{51}1$$
$$=\frac{1}{4}\times 51^2\times 52^2+147\times\frac{51}{6}\times 52\times 103+7203\times\frac{51}{2}\times 52+117648\times 51$$
$$=24\,001\,824$$

28)

$$51^2+52^2+53^2+\cdots+101^2=\sum_{r=51}^{101}r^2$$
$$=\sum_{r=1}^{51}(r+50)^2=\sum_{r=1}^{51}(r^2+100r+2500)$$
$$=\sum_{r=1}^{51}r^2+100\sum_{r=1}^{51}r+2500\sum_{r=1}^{51}1$$
$$=\frac{1}{6}\times 51(51+1)(102+1)+100\times\frac{1}{2}\times 51(52)+2500\times 51$$
$$=305\,626$$

29 a)

$$\sum_{r=1}^{n} r(r-1)(r+1) = \sum_{r=1}^{n} r(r^2-1) = \sum_{r=1}^{n} (r^3 - r)$$
$$= \sum_{r=1}^{n} r^3 - \sum_{r=1}^{n} r$$
$$= \frac{1}{4}n^2(n+1) - \frac{1}{2}n(n+1)$$
$$= \frac{1}{4}n(n+1)[n(n+1) - 2]$$
$$= \frac{1}{4}n(n+1)[n^2 + n - 2]$$
$$= \frac{1}{4}n(n+1)(n-1)(n+2) \text{ as req.}$$

29 b)

$$\sum_{r=1}^{2n} r(r-1)(r+1) = \frac{1}{4} \times 2n(2n+1)(2n-1)(2n+2)$$
$$= \frac{1}{2} \times 2n(2n+1)(2n-1)(n+1)$$
$$= n(2n+1)(2n-1)(n+1) \text{ as req.}$$

30 a)

$$\sum_{r=1}^{2n} r^3 = \frac{1}{4}(2n)^2(2n+1)^2 = n^2(2n+1)^2$$

30 b)

$$\sum_{r=n+1}^{2n} r^2 = \sum_{r=1}^{2n} r^2 - \sum_{r=1}^{n} r^2$$
$$= \frac{1}{6}(2n)(2n+1)(2(2n)+1) - \frac{1}{6}n(n+1)(2n+1)$$
$$= \frac{1}{3}n(2n+1)(4n+1) - \frac{1}{6}n(n+1)(2n+1)$$
$$= \frac{1}{6}n[2(2n+1)(4n+1) - (n+1)(2n+1)]$$
$$= \frac{1}{6}n[16n^2 + 12n + 2 - 2n^2 - 3n - 1]$$
$$= \frac{1}{6}n(14n^2 + 9n + 1)$$
$$= \frac{1}{6}n(7n+1)(2n+1)$$

Alternative Method:

$$\sum_{r=n+1}^{2n} r^2 = \sum_{r=1}^{n}(r+n)^2$$
$$= \sum_{r=1}^{n}(r^2+2nr+n^2)$$
$$= \frac{1}{6}n(n+1)(2n+1) + 2n \times \frac{1}{2}n(n+1) + n^3$$
$$= \frac{1}{6}n[(n+1)(2n+1) + 6n(n+1) + 6n^2]$$
$$= \frac{1}{6}n[2n^2+3n+1+6n^2+6n+6n^2]$$
$$= \frac{1}{6}n(14n^2+9n+1)$$
$$= \frac{1}{6}n(7n+1)(2n+1)$$

30 c)

$$\sum_{r=1}^{n+4} r^2 = \frac{1}{6}(n+4)(n+4+1)(2(n+4)+1)$$
$$= \frac{1}{6}(n+4)(n+5)(2n+9)$$

30 d)

$$\sum_{r=1}^{n+3}(r^3-2r^2) = \sum_{r=1}^{n+3} r^3 - 2\sum_{r=1}^{n+3} r^2$$
$$= \frac{1}{4}(n+3)^2(n+3+1)^2 - 2 \times \frac{1}{6}(n+3)(n+3+1)(2(n+3)+1)$$
$$= \frac{1}{4}(n+3)^2(n+4)^2 - \frac{1}{3}(n+3)(n+4)(2n+7)$$
$$= \frac{1}{12}(n+3)(n+4)[3(n+3)(n+4) - 4(2n+7)]$$
$$= \frac{1}{12}(n+3)(n+4)[3n^2+21n+36-8n-28]$$
$$= \frac{1}{12}(n+3)(n+4)(3n^2+13n+8)$$

30 e)

$$\sum_{r=n+1}^{2n}(r^3+r)=\sum_{r=1}^{2n}(r^3+r)-\sum_{r=1}^{n}(r^3+r)$$
$$=\sum_{r=1}^{2n}r^3+\sum_{r=1}^{2n}r-\sum_{r=1}^{n}r^3-\sum_{r=1}^{n}r$$
$$=\frac{1}{4}(2n)^2(2n+1)^2+\frac{1}{2}(2n)((2n)+1)-\frac{1}{4}n^2(n+1)^2-\frac{1}{2}n(n+1)$$
$$=n^2(2n+1)^2+n(2n+1)-\frac{1}{4}n^2(n+1)^2-\frac{1}{2}n(n+1)$$
$$=\frac{1}{4}n[4n(2n+1)^2+4(2n+1)-n(n+1)^2-2(n+1)]$$
$$=\frac{1}{4}n[4n(4n^2+4n+1)+4(2n+1)-n(n^2+2n+1)-2(n+1)]$$
$$=\frac{1}{4}n[16n^3+16n^2+4n+8n+4-n^3-2n^2-n-2n-2]$$
$$=\frac{1}{4}n(15n^3+14n^2+9n+2)$$
$$=\frac{1}{4}n(3n+1)(5n^2+3n+2)$$

Alternative Method:

$$\sum_{r=n+1}^{2n}(r^3+r)=\sum_{r=1}^{n}((r+n)^3+(r+n))$$
$$=\sum_{r=1}^{n}(r^3+3nr^2+3n^2r+r+n+n^3)$$
$$=\frac{1}{4}n^2(n+1)^2+3n\times\frac{1}{6}n(n+1)(2n+1)+3n^2\times\frac{1}{2}n(n+1)$$
$$+\frac{1}{2}n(n+1)+n^2+n^4$$
$$=n\left[\frac{1}{4}n(n+1)^2+\frac{1}{2}n(n+1)(2n+1)+\frac{3}{2}n^2(n+1)+\frac{1}{2}(n+1)+n+n^3\right]$$
$$=n\left[\frac{1}{4}n(n^2+2n+1)+\frac{1}{2}n(2n^2+3n+1)+\frac{3}{2}(n^3+n^2)+\frac{1}{2}(n+1)+n+n^3\right]$$
$$=n\Bigg[\frac{1}{4}n^3+\frac{1}{2}n^2+\frac{1}{4}n+n^3+\frac{3}{2}n^2+\frac{1}{2}n$$
$$+\frac{3}{2}n^3+\frac{3}{2}n^2+\frac{1}{2}n+\frac{1}{2}+n+n^3\Bigg]$$
$$=n\left[\frac{15}{4}n^3+\frac{7}{2}n^2+\frac{9}{4}n+\frac{1}{2}\right]$$
$$=\frac{1}{4}n(15n^3+14n^2+9n+2)$$
$$=\frac{1}{4}n(3n+1)(5n^2+3n+2)$$

31)

$$\sum_{r=6}^{n+1} r^2 = \sum_{r=1}^{n+1} r^2 - \sum_{r=1}^{5} r^2$$

$$= \frac{1}{6}(n+1)(n+1+1)(2(n+1)+1) - \frac{1}{6} \times 5(5+1)(10+1)$$

$$= \frac{1}{6}(n+1)(n+2)(2n+3) - 55$$

$$= \frac{1}{6}[(n+1)(2n^2+7n+6) - 330]$$

$$= \frac{1}{6}[2n^3+9n^2+13n+6-330]$$

$$= \frac{1}{6}(2n^3+9n^2+13n-324)$$

32 a)

$$\sum_{r=1}^{n} r(r-5) = \sum_{r=1}^{n} r^2 - 5r = \sum_{r=1}^{n} r^2 - 5\sum_{r=1}^{n} r$$

$$= \frac{1}{6}n(n+1)(2n+1) - 5 \times \frac{1}{2}n(n+1)$$

$$= \frac{1}{6}n[(n+1)(2n+1) - 15(n+1)]$$

$$= \frac{1}{6}n[2n^2+3n+1-15n-15]$$

$$= \frac{1}{6}n(2n^2-12n-14)$$

$$= \frac{1}{3}n(n^2-6n-7)$$

$$= \frac{1}{3}n(n-7)(n+1)$$

32 b)

$$\sum_{r=n+1}^{3n} r(r-5) = \sum_{r=1}^{3n} r(r-5) - \sum_{r=1}^{n} r(r-5)$$

$$= \frac{1}{3}(3n)(3n-7)(3n+1) - \frac{1}{3}n(n-7)(n+1)$$

$$= \frac{1}{3}n[3(3n-7)(3n+1) - (n-7)(n+1)]$$

$$= \frac{1}{3}n[3(9n^2-18n-7) - (n^2-6n-7)]$$

$$= \frac{1}{3}n(27n^2-54n-21-n^2+6n+7)$$

$$= \frac{2}{3}n(13n^2-24n-7)$$

33 a)

$$\sum_{r=1}^{n}(2r)^2 = 4\sum_{r=1}^{n} r$$
$$= 4 \times \frac{1}{6}n(n+1)(2n+1) = \frac{2}{3}n(n+1)(2n+1)$$

33 b)

$$\sum_{r=k}^{k+4}(2r)^2 = 1020$$
$$\sum_{r=k}^{k+4}(2r)^2 = \sum_{r=1}^{k+4}(2r)^2 - \sum_{r=1}^{k-1}(2r)^2$$
$$= \frac{2}{3}(k+4)(k+4+1)(2(k+4)+1)$$
$$-\frac{2}{3}(k-1)(k-1+1)(2(k-1)+1)$$
$$\frac{2}{3}(k+4)(k+5)(2k+9) - \frac{2}{3}k(k-1)(2k-1) = 1020$$
$$(k+4)(k+5)(2k+9) - k(k-1)(2k-1) = 1530$$
$$(k+4)(2k^2+19k+45) - k(2k^2-3k+1) = 1530$$
$$2k^3+19k^2+45k+8k^2+76k+180-2k^3+3k^2-k = 1530$$
$$30k^2+120k-1350 = 0$$

which has solutions $k = -9$ or $k = 5$

Since k is positive, $k = 5$ so $2k = 10$

34)

$$\sum_{r=1}^{2n} r^3 = \frac{1}{4}(2n)^2(2n+1)^2 = n^2(4n^2+4n+1) = n(4n^3+4n^2+n)$$

$$\sum_{r=1}^{n}(r+2)^2 = \sum_{r=1}^{n}(r^2+4r+4)$$

$$= \sum_{r=1}^{n} r^2 + 4\sum_{r=1}^{n} r + 4\sum_{r=1}^{n} 1$$

$$= \frac{1}{6}n(n+1)(2n+1) + 4 \times \frac{1}{2}n(n+1) + 4n$$

$$= \frac{1}{6}n[(n+1)(2n+1) + 12(n+1) + 24]$$

$$= \frac{1}{6}n[2n^2+3n+1+12n+12+24]$$

$$= \frac{1}{6}n(2n^2+15n+37)$$

$$\sum_{r=1}^{2n} r^3 - \sum_{r=1}^{n}(r+2)^2 = n(4n^3+4n^2+n) - \frac{1}{6}n(2n^2+15n+37)$$

$$= \frac{1}{6}n[24n^3+24n^2+6n-2n^2-15n-37]$$

$$= \frac{1}{6}n[24n^3+22n^2-9n-37]$$

$$= \frac{1}{6}n(n-1)(24n^2+46n+37)$$

35)

$$\sum_{r=1}^{n}(r+a)(r+b) = \sum_{r=1}^{n}(r^2+(a+b)r+ab)$$

$$= \sum_{r=1}^{n} r^2 + (a+b)\sum_{r=1}^{n} r + ab\sum_{r=1}^{n} 1$$

$$= \frac{1}{6}n(n+1)(2n+1) + (a+b) \times \frac{1}{2}n(n+1) + abn$$

$$= \frac{1}{3}n\left[\frac{1}{2}(n+1)(2n+1) + \frac{3}{2}(a+b)(n+1) + 3ab\right]$$

$$= \frac{1}{3}n\left[n^2 + \frac{3}{2}n + \frac{1}{2} + \frac{3}{2}(a+b)n + \frac{3}{2}(a+b) + 3ab\right]$$

$$= \frac{1}{3}n\left[n^2 + \left(\frac{3}{2} + \frac{3}{2}(a+b)\right)n + \frac{1}{2} + \frac{3}{2}(a+b) + 3ab\right]$$

So comparing terms with $\frac{1}{3}n(n^2-7)$

$$\frac{3}{2} + \frac{3}{2}(a+b) = 0$$

$$\frac{1}{2} + \frac{3}{2}(a+b) + 3ab = -7$$

Rearrange the first:

$$\frac{3}{2}(a+b) = -\frac{3}{2} \Rightarrow a = -b-1$$

Sub into second:

$$\frac{1}{2}+\frac{3}{2}(-b-1+b)+3(-b-1)b=-7$$
$$-1-3b^2-3b=-7$$

$3b^2+3b-6=0$ which has solutions $b=-2$ or $b=1$

$\therefore b=-2$, $a=1$ or $a=-2$, $b=1$

36 a)

$$\sum_{r=1}^{2n}(n-r)^2=\sum_{r=1}^{2n}(n^2-2nr+r^2)$$
$$=n^2\sum_{r=1}^{2n}1-2n\sum_{r=1}^{2n}r+\sum_{r=1}^{2n}r^2$$
$$=n^2(2n)-2n\times\frac{1}{2}(2n)(2n+1)+\frac{1}{6}(2n)(2n+1)(4n+1)$$
$$=2n^3-2n^2(2n+1)+\frac{1}{3}n(2n+1)(4n+1)$$
$$=\frac{1}{3}n[6n^2-6n(2n+1)+(2n+1)(4n+1)]$$
$$=\frac{1}{3}n[6n^2-12n^2-6n+8n^2+6n+1]$$
$$=\frac{1}{3}n(2n^2+1)\text{ as req.}$$

36 b)

$$\sum_{r=1}^{n}(n-r)^2=n^2\sum_{r=1}^{n}1-2n\sum_{r=1}^{n}r+\sum_{r=1}^{n}r^2$$
$$=n^2\times n-2n\times\frac{1}{2}n(n+1)+\frac{1}{6}n(n+1)(2n+1)$$
$$=\frac{1}{6}n[6n^2-6n(n+1)+(n+1)(2n+1)]$$
$$=\frac{1}{6}n[6n^2-6n^2-6n+2n^2+3n+1]$$
$$=\frac{1}{6}n(2n^2-3n+1)$$

So that

$$\sum_{r=n+1}^{2n}(n-r)^2=\sum_{r=1}^{2n}(n-r)^2-\sum_{r=1}^{n}(n-r)^2$$
$$=\frac{1}{3}n(2n^2+1)-\frac{1}{6}n(2n^2-3n+1)$$
$$=\frac{1}{6}n[2(2n^2+1)-2n^2+3n-1]$$
$$=\frac{1}{6}n[4n^2+2-2n^2+3n-1]$$
$$=\frac{1}{6}n[2n^2+3n+1]$$
$$=\frac{1}{6}n(n+1)(2n+1)=\sum_{r=1}^{n}r^2$$

as req.

37 a)

$$\sum_{r=1}^{3n-2}(5r+3) = 5\sum_{r=1}^{3n-2} r + \sum_{r=1}^{3n-2} 3$$

$$= 5\left(\frac{1}{2}(3n-2)(3n-2+1)\right) + 3(3n-2)$$

$$= \frac{5}{2}(9n^2 - 9n + 2) + 9n - 6$$

$$= \frac{1}{2}(45n^2 - 45n + 10) + \frac{1}{2}(18n - 12)$$

$$= \frac{1}{2}(45n^2 - 27n - 2)$$

$$= \frac{1}{2}(3n-2)(15n+1)$$

37 b)

$$3n - 2 = 16 \Rightarrow n = 6$$

$$\frac{1}{2}(45 \times 6^2 - 27 \times 6 - 2) = 728$$

38 a)

$$\sum_{r=1}^{2n+7}(r^2 - 1) = \sum_{r=1}^{2n+7} r^2 - \sum_{r=1}^{2n+7} 1$$

$$= \frac{1}{6}(2n+7)(2n+7+1)(2(2n+7)+1) - (2n+7)$$

$$= \frac{1}{6}(2n+7)(2n+8)(4n+15) - (2n+7)$$

$$= \frac{1}{3}(2n+7)[(n+4)(4n+15) - 3]$$

$$= \frac{1}{3}(2n+7)[4n^2 + 31n + 57]$$

$$= \frac{1}{3}(2n+7)(4n+19)(n+3) \text{ as req.}$$

38 b)

$$2n + 7 = 63 \Rightarrow n = 28$$

$$\frac{1}{3}(2 \times 28 + 7)(4 \times 28 + 19)(28 + 3) = 85\,281$$

$$2n + 7 = 9 \Rightarrow n = 1$$

$$\frac{1}{3}(2 \times 1 + 7)(4 \times 1 + 19)(1 + 3) = 276$$

$$85\,281 - 276 = 85\,005$$

39 a)

$$\sum_{r=1}^{8}(-1)^r = -1+1-1+1-1+1-1+1$$
$$= 4(-1)+4(1) = 0$$

39 b)

$$\sum_{r=1}^{2n}(-1)^r = -1+1-\cdots-1+1 = n(-1)+n(1) = 0$$

39 c)

$$\sum_{r=1}^{9}(-1)^r = -1+1-1+1-1+1-1+1-1$$
$$= 5(-1)+4(1) = -1$$

39 d)

$$\sum_{r=1}^{2n-1}(-1)^r = -1+1-\cdots-1+1-1$$
$$= n(-1)+(n-1)(1) = -1$$

40 a)

$$\sum_{r=1}^{2n}(-1)^r r = -\sum_{r=1}^{n}(2r-1)+\sum_{r=1}^{n}(2r)$$
$$= -2\sum_{r=1}^{n} r+\sum_{r=1}^{n}1+2\sum_{r=1}^{n} r$$
$$= \sum_{r=1}^{n}1 = n$$

40 b)

$$\sum_{r=1}^{2n-1}(-1)^r r = -\sum_{r=1}^{n}(2r-1)+\sum_{r=1}^{n-1}(2r)$$
$$= -2\sum_{r=1}^{n} r+\sum_{r=1}^{n}1+2\sum_{r=1}^{n-1} r$$
$$= -2\times\frac{1}{2}n(n+1)+n+2\times\frac{1}{2}(n-1)n$$
$$= -n^2-n+n+n^2-n$$
$$= -n$$

40 c)

$$\begin{aligned}\sum_{r=1}^{2n}(-1)^r(r+1) &= -\sum_{r=1}^{n}(2r)+\sum_{r=1}^{n}(2r+1)\\ &= -2\sum_{r=1}^{n}r+2\sum_{r=1}^{n}r+\sum_{r=1}^{n}1\\ &= n\end{aligned}$$

40 d)

$$\begin{aligned}\sum_{r=1}^{2n}(-1)^r r^2 &= -\sum_{r=1}^{n}(2r-1)^2+\sum_{r=1}^{n}(2r)^2\\ &= -4\sum_{r=1}^{n}r^2+4\sum_{r=1}^{n}r-\sum_{r=1}^{n}1+4\sum_{r=1}^{n}r^2\\ &= 4\sum_{r=1}^{n}r-\sum_{r=1}^{n}1\\ &= 4\times\frac{1}{2}n(n+1)-n\\ &= 2n^2+2n-n\\ &= n(2n+1)\end{aligned}$$

41 a)

$$\begin{aligned}\sum_{r=1}^{10}\sin\left(\frac{\pi}{2}r\right) &= \sin\frac{\pi}{2}+\sin\pi+\sin\frac{3\pi}{2}+\sin 2\pi+\cdots+\sin 5\pi\\ &= 1+0-1+0+1+0-1+0+1+0\\ &= 1\end{aligned}$$

41 b)

$$\begin{aligned}&\sum_{r=1}^{32}r\cos\left(\frac{\pi}{2}r\right)\\ &= \cos\frac{\pi}{2}+2\cos\pi+3\cos\frac{3\pi}{2}+4\cos 2\pi+\cdots+32\cos 5\pi\\ &= 0-2+0+4+0-6+0+1+\cdots+0-30+0+32\\ &= -2+4-6+8+\cdots-30+32\\ &= 8\times 2=16\end{aligned}$$

41 c)

$$\sum_{r=1}^{27} r\tan\left(\frac{\pi}{3}r\right)$$
$$= \tan\frac{\pi}{3} + 2\tan\frac{2\pi}{3} + 3\tan\pi + 4\tan\frac{4\pi}{3} + \cdots + 27\tan\pi$$
$$= \sqrt{3} - 2\sqrt{3} + 0 + 4\sqrt{3} - 6\sqrt{3} + \cdots + 25\sqrt{3} - 26\sqrt{3} + 0$$
$$= 9 \times -\sqrt{3} = -9\sqrt{3}$$

42 a)

$$\sum_{r=1}^{n} \sin(\pi r) = \sin\pi + \sin 2\pi + \sin 3\pi + \cdots + \sin n\pi$$
$$= 0$$

42 b)

$$\sum_{r=1}^{n} \cos(2\pi r) = \cos(2\pi) + \cos(4\pi) + \cos(6\pi) + \cdots + \cos(2\pi n)$$
$$= 1 + 1 + 1 + \cdots + 1$$
$$= n$$

42 c)

$$\sum_{r=1}^{3n} \tan\left(\frac{\pi}{3}r\right) = \tan\left(\frac{\pi}{3}\right) + \tan\left(\frac{2\pi}{3}\right) + \tan(\pi) + \cdots + \tan(n\pi)$$
$$= \sqrt{3} - \sqrt{3} + 0 + \cdots + \sqrt{3} - \sqrt{3} + 0$$
$$= 0$$

43 a)

$$\sum_{r=1}^{n}(2r+1)^2 = \sum_{r=1}^{n}(4r^2 + 4r + 1)$$
$$= 4\sum_{r=1}^{n} r^2 + 4\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1$$
$$= 4 \times \frac{1}{6}n(n+1)(2n+1) + 4 \times \frac{1}{2}n(n+1) + n$$
$$= \frac{1}{3}n[2(n+1)(2n+1) + 6(n+1) + 3]$$
$$= \frac{1}{3}n[4n^2 + 6n + 2 + 6n + 6 + 3]$$
$$= \frac{1}{3}n(4n^2 + 12n + 11) \text{ as req.}$$

43 b)

$$\sum_{r=1}^{k} \cos(2\pi r) = 1 + 1 + 1 + \cdots + 1 = k$$

$$\therefore 15\sum_{r=1}^{k} \cos(2\pi r) = 15k$$

$$\therefore \sum_{r=1}^{k} (2r+1)^2 + 15\sum_{r=1}^{k} \cos(2\pi r) = \frac{1}{3}k(4k^2 + 12k + 11) + 15k$$

$$\frac{1}{3}k(4k^2 + 12k + 11) + 15k = 24k^2 \Rightarrow \frac{4}{3}k^3 - 20k^2 + \frac{56}{3}k = 0$$

Which gives $k = 0$, $k = 1$ or $k = 14$

So $k = 1$ or $k = 14$

44 a)

$$\sum_{r=1}^{n} (5r-3)^2 = \sum_{r=1}^{n} (25r^2 - 30r + 9)$$

$$= 25\sum_{r=1}^{n} r^2 - 30\sum_{r=1}^{n} r + 9\sum_{r=1}^{n} 1$$

$$= 25 \times \frac{1}{6}n(n+1)(2n+1) - 30 \times \frac{1}{2}n(n+1) + 9n$$

$$= \frac{1}{6}n[25(n+1)(2n+1) - 90(n+1) + 54]$$

$$= \frac{1}{6}n[50n^2 + 75n + 25 - 90n - 90 + 54]$$

$$= \frac{1}{6}n(50n^2 - 15n - 11) \text{ as req.}$$

44 b)

$$\sum_{r=1}^{20} r\sin\left(\frac{\pi}{2}r\right)$$

$$= 1 \times 1 + 2 \times 0 + 3 \times -1 + 4 \times 0 + \cdots + 17 \times 1 + 18 \times 0 + 19 \times -1 + 20 \times 0$$

$$= 1 - 3 + \cdots + 17 - 19 = 5 \times -2 = -10$$

$$\therefore 20\sum_{r=1}^{20} r\sin\left(\frac{\pi}{2}r\right) = 20 \times -10$$

$$\sum_{r=3}^{k} (5r-3)^2 = \frac{1}{6}k(50k^2 - 15k - 11) - 4 - 49$$

$$= \frac{25}{3}k^3 - \frac{5}{2}k^2 - \frac{11}{6}k - 53$$

$$\therefore \sum_{r=3}^{k} (5r-3)^2 + 20\sum_{r=1}^{20} r\sin\left(\frac{\pi}{2}r\right) = \frac{25}{3}k^3 - \frac{5}{2}k^2 - \frac{11}{6}k - 253$$

$$\frac{25}{3}k^3 - \frac{5}{2}k^2 - \frac{11}{6}k - 253 = 241k \Rightarrow \frac{25}{3}k^3 - \frac{5}{2}k^2 - \frac{1457}{6}k - 253 = 0$$

Which gives $k = -\frac{23}{5}$, $k = -\frac{11}{10}$ or $k = 6$ So $k = 6$

45 a)

$$\begin{aligned}
\sum_{r=1}^{2n} r((-1)^r+5)^2 &= \sum_{r=1}^{2n} r(1+(-1)^r 10+25) \\
&= \sum_{r=1}^{2n}(26r+(-1)^r 10r) \\
&= 26\sum_{r=1}^{2n} r - 10\sum_{r=1}^{n}(2r-1) + 10\sum_{r=1}^{n}(2r) \\
&= 26\sum_{r=1}^{2n} r - 20\sum_{r=1}^{n} r + 10\sum_{r=1}^{n} 1 + 20\sum_{r=1}^{n} r \\
&= 26\sum_{r=1}^{2n} r + 10\sum_{r=1}^{n} 1 \\
&= 26\times\frac{1}{2}(2n)(2n+1) + 10n \\
&= 26n(2n+1) + 10n \\
&= 52n^2 + 26n + 10n \\
&= 52n^2 + 36n \\
&= n(52n+36)
\end{aligned}$$

45 b)

$$40(52\times 40+36) - 7(52\times 7+36) = 81\,840$$

46 a)

$$\begin{aligned}
\sum_{r=1}^{2n}((-1)^r+3r)^2 &= \sum_{r=1}^{2n}(1+(-1)^r 6r+9r^2) \\
&= \sum_{r=1}^{2n} 1 - 6\sum_{r=1}^{n}(2r-1) + 6\sum_{r=1}^{n}(2r) + 9\sum_{r=1}^{2n} r^2 \\
&= \sum_{r=1}^{2n} 1 - 12\sum_{r=1}^{n} r + 6\sum_{r=1}^{n} 1 + 12\sum_{r=1}^{n} r + 9\sum_{r=1}^{2n} r^2 \\
&= \sum_{r=1}^{2n} 1 + 6\sum_{r=1}^{n} 1 + 9\sum_{r=1}^{2n} r^2 \\
&= 2n + 6n + 9\times\frac{1}{6}\times 2n(2n+1)(2(2n)+1) \\
&= 2n + 6n + 3n(2n+1)(4n+1) \\
&= 2n + 6n + 3n(8n^2+6n+1) \\
&= 2n + 6n + 24n^3 + 18n^2 + 3n \\
&= 24n^3 + 18n^2 + 11n
\end{aligned}$$

46 b)

$$\begin{aligned}
&24\times 50^3 + 18\times 50^2 + 11\times 50 - (24\times 24^3 + 18\times 24^2 + 11\times 24) \\
&= 2\,703\,142
\end{aligned}$$